

AMANDA AULIA

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About Me

Top Graduate Bachelor's Degree in Data Science at Universitas Pembangunan Nasional "Veteran" Jawa Timur (GPA 3.89/4.00) and **Distinction Graduate of Bangkit Academy 2023 Batch 2**. Strong background in **data analysis, statistics, and machine learning**, with hands-on experience using **Python, SQL, Excel, Tableau, and SPSS**. Actively participated in professional training programs, student organizations, and committees at both university and external levels. Currently seeking professional opportunities to apply analytical skills and gain practical experience in the data field. Possess strong **communication, teamwork, and problem-solving skills**, with the ability to work effectively both independently and collaboratively. Highly motivated to continuously learn and grow as a **Data Analyst** and **Data Scientist**.

Education

Universitas Pembangunan Nasional Veteran Jawa Timur – Surabaya September 2021 - December 2025

Bachelor's Degree – Data Science, Graduated with honors (Cum Laude), GPA 3.89/4.00

- Best Graduates of the Data Science Study Program at the 97th UPN Veteran Jawa Timur Graduation Ceremony.
- Relevant Coursework: Programming Algorithms, Database, Mathematics, Statistic, Machine Learning, Data Mining, Big Data, Data Analytics

Bangkit Academy 2023 Batch 2 August 2023 - January 2024

Non-Degree (Study Independent) – Machine Learning Path, Distinction Graduate with score 93.98/100

- Completed 11 intensive courses covering Python, SQL, Data Storytelling, Data Analysis, Machine Learning, Deep Learning, Computer Vision, and Tensorflow.
- Contributed to a collaborative capstone project developing "Laper.in", an Android app aimed at addressing food waste in Indonesia. Involved in developing the personalized recipe recommendation system and ingredient classification model, contributing to innovative sustainability solutions
- Top 1000 most active student in Instructor-Led Training and achieved a score of 98% in an English for Business Communication

Experience

Dinas Sosial Provinsi Jawa Timur – Surabaya February 2024 – June 2024

Data Analyst Intern – Disaster Management Division

- Developed a disaster monitoring information system website for East Java using dashboards and spatial maps, implemented with Flask and MySQL, including daily disaster event data entry.
- Conducted exploratory data analysis and applied clustering techniques to provide recommendations based on personnel data, particularly in disaster response units.
- Performed rainfall time series analysis using the ARIMA method.
- Analyzed Instagram social media data of the East Java Provincial Social Service and created data visualizations.

Publication

FORRISX: Forecasting and Risk Analysis System Using ARIMAX and VaR 2026

Published in Directorate General of Intellectual Property Right (DJKI) under the Ministry of Law and Human Rights of the Public of Indonesia (Kemenkumham RI)

- Developed a predictive analytics web system for stock price forecasting using the ARIMAX time series method.
- Implemented the Value-at-Risk (VaR) method in a web-based system to measure and visualize potential investment loss risk

Forecasting Financial Sector Stock Price and Loss Risk Using the ARIMAX and Value-at-Risk Methods 2025

Published in Journal Binary Digital – Technology (BIT-TECH)

- Applied an ARIMAX model integrated with macroeconomic variables (USD/IDR and SGD/IDR) to forecast BBKA stock prices for the 2019–2024 period, achieving high accuracy with a MAPE of 2.19%.
- Integrated forecasting and investment risk management using the Value-at-Risk (VaR) method with 95% historical simulation confidence level, estimating a potential daily loss of 1.71% to support investment decision-making.

Topic-Based Legislation Network Mapping with Chatgpt and NEO4J 2024

Published in International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS)

- Developed a web-based Legal Network Analytics (LNA) system to map relationships between laws and regulations based on topic similarity.
- Implemented Natural Language Processing (NLP) using GPT-3.5-Turbo and Neo4j graph databases to extract regulatory topics and visualize legal networks interactively.

Funding Awardee Program Kreativitas Mahasiswa Karsa Cipta (PKM-KC) 2023 “FishGrom: Alat Monitoring Perkembangan Ikan Non-Kontak Berbasis *Binocular Stereo Vision* untuk Mendeteksi Jumlah dan Ukuran Ikan Bandeng pada Tambak” 2023

Published in Directorate General of Intellectual Property Right (DJKI) under the Ministry of Law and Human Rights of the Public of Indonesia (Kemenkumham RI)

- Conducted in-depth analysis of system methodologies, designed system architecture, prepared project budgets, and developed proposals following proper academic writing standards.
- Implemented a contour-based object detection system using OpenCV with an error rate of only 0.21%.
- The proposal received IDR 9,000,000 in funding from the 2023 Student Creativity Program (PKM) and was officially registered with DJKI, obtaining a copyright certificate.

Information Retrieval of Study Program Activity News Using the TF-IDF Weighting Algorithm and Cosine Similarity 2022

Published in Seminar Nasional Teknologi dan Sistem Informasi (SITASI)

- Developed a prototype information retrieval system for searching study program news content using TF-IDF weighting and Cosine Similarity algorithms.
- Testing on 22 Data Science study program news articles at UPN “Veteran” Jawa Timur achieved 100% research accuracy.

Activity & Projects

Capstone Project Bangkit 2023 Batch 2 “Laper.in: Make Food with Several Recipe” 2023

- Collected a food ingredient image dataset consisting of 15 ingredient classes, with 450 training images, 25 testing images, and 25 validation images.
- Compiled food recipes, preparation instructions, and tutorial video links.
- Built an image classification model using a pre-trained MobileNetV2 model and deployed the output in TFLite format.

Principal Component Analysis (PCA) Dimensionality Reduction Impact Analysis 2023

- Evaluated the impact of dimensionality reduction using PCA on classification model performance using a hepatitis patient dataset with 155 records and 20 features.
- Identified superior performance of Decision Tree before PCA (accuracy 90.3%, F1-score 95.8%, specificity 85.7%) and performance improvement in Logistic Regression after PCA (accuracy 90.3%, F1-score 93.3%, specificity 71.4%).
- Concluded that PCA improved Logistic Regression performance while reducing Decision Tree effectiveness.

Classification Model Performance Comparison Analysis 2023

- Conducted performance comparison of K-Nearest Neighbor, Logistic Regression, Naïve Bayes, and Decision Tree models using the Iris and Patient Treatment datasets.
- The process included data collection, preprocessing (cleaning and transformation), exploratory data analysis, feature engineering, model development, and performance evaluation.
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Organization Experience

Badminton Student Activity Unit – UPN Veteran Jawa Timur 2024-2025

Treasurer

- Develop a budget plan to meet the needs of the event, oversee the income and use of funds.
- Coordinate with other division regarding funding needs and create detailed and clear funding books.
- Make an accountability report for financial management.

Badminton Student Activity Unit – UPN Veteran Jawa Timur 2023-2024

Public Relation Departement

- Managing communications and publications related to the activities of the badminton student activity unit for internal and external parties
- Serving as a contact person and liaison for delivery of information, invitation, and activity collaborations.

Data Science Student Association (HIMASADA) – UPN Veteran Jawa Timur 2023-2024

Staff of Commission 2 of The Legislature Department (BLJ)

- Present the result of the aspiration net at each public hearing meeting
- Monitoring the performance of the executive department (BEJ) and adjusting the assessment of key performance indicators and submitting them to commission 1

Data Science Student Association (HIMASADA) – UPN Veteran Jawa Timur 2022-2023

Student Advocacy and Welfare Department (ADVOKESMA)

- Providing advocacy and welfare services to all Data Science student in matters related to academic affairs such as registration, study plan, UKT appeal requests and help new student at the beginning of admission.
- Distributing information related to policies from the university, faculty, and study program to Data Science student

Committee Experiences

• Veteran Badminton Competition 2024 – Badminton Student Activity Unit <i>Steering Committee (SC)</i>	2024
• Malam Keakraban 2024 – Badminton Student Activity Unit <i>Steering Committee (SC)</i>	2024
• Sambang Desa 2023 – Data Science Student Association (HIMASADA) <i>Volunteer & Best Mentor</i>	2023
• Malam Keakraban 2023 – Badminton Student Activity Unit <i>Treasurer</i>	2023
• Orientasi Mahasiswa Sains Data (OSADA) 2023 – Data Science Student Association (HIMASADA) <i>Secretary</i>	2023
• Akshaya 2022 – Data Science Student Association (HIMASADA) <i>Security Division</i>	2022
• Sains Data E-Sport Competition (SESCOP) – Data Science Student Association (HIMASADA) <i>Event Division</i>	2022
• Malam Keakraban 2022 – Badminton Student Activity Unit <i>Equipment Division</i>	2022
• Ngobrol Seputar Kehidupan Sains Data (NGOSA KATA) – Data Science Student Association (HIMASADA) <i>Secretary</i>	2022
• Kontes Kapal Cepat Tak Berawak Nasional (KKCTBN) 2022 – Pusat Prestasi Nasional <i>Liaison Officer Division</i>	2022
• Orientasi Mahasiswa Sains Data (OSADA) 2022 – Data Science Student Association (HIMASADA) <i>Disciplinary Commission Division</i>	2022
• Dialog Internal Sains Data (LOGISTA) 2022 – Data Science Student Association (HIMASADA) <i>Equipment Division</i>	2022

Certificates & Training

- Coursera – TensorFlow: Data and Deployment Specialization (November 2023)
- Coursera – DeepLearning.AI TensorFlow Developer Specialization (November 2023)
- Coursera – Machine Learning Specialization (November 2023)
- Coursera – Mathematics for Machine Learning and Data Science Specialization (September 2023)
- Dicoding – Memulai Dasar Pemrograman untuk Menjadi Pengembang Software (Agustus 2023)
- Dicoding – Pengenalan ke Logika Pemrograman (Programming Logic 101) (Agustus 2023)
- Dicoding – Belajar Dasar Git dengan Github
- Badan Nasional Sertifikasi Profesi (BNSP) – Junior Web Programmer (April 2023)
- Sains Data UPN “Veteran” Jawa Timur – Pelatihan “Data Bercerita Menggunakan Tableau” (Juli 2022)
- Sains Data UPN “Veteran” Jawa Timur – Pelatihan “Pengelolaan dan Analisis Data dengan SPSS” (Juli 2022)

Skills & Competencies

Data Analytics

- Data Analysis, Data Mining, Data Visualization, Data Wrangling, Machine Learning

Programming Language

- Python, R, SQL

Tools

- SPSS, Tableau, MySQL, Google Colab, Visual Studio Code, Excel/Spreadsheet, Word

Supporting Skills

- Analytical Thinking, Problem-Solving, Effective Communication, Collaborative Teamwork

Language

- Indonesia (Native), English (Professional Proficiency)